Hacettepe University

Department of Computer Engineering

CMP634 Computer Architecture

MIPS Project Assignment

Deadline: Sunday, 01/11/2026 (at 23:59)
Submission Form: `https://docs.google.com/forms/d/e/1FAIpQLSeS8Mobm9Gg5LrGt4q__PNrrO5mDeC2fJZFQCaKF96HXEwDJA/viewform?usp=header`

Overview

In this project, you will write and simulate MIPS assembly code using either the **QtSpim** or **MARS** simulator. Please download your preferred simulator and familiarize yourself with its usage before starting.

Questions

1) Arrays with Loops (50 Points)

Assume you have an array with 10 integer elements (e.g., $A[10]$) and an integer value K. Write a MIPS assembly program that counts and returns the total number of elements in A that are **both strictly smaller than** K ($A[i] < K$) and **even** ($A[i] \pmod 2 == 0$).

Example: Assume $A = \{2, 13, 1, 4, 8, 20, 5, 6, 9, 14\}$.

- For $K = 10$, your program should return 3 (the valid elements are $2, 4$, and 6).
- For $K = 20$, your program should return 5 (the valid elements are $2, 4, 8, 6$, and 14).

Array Definition and Base Skeleton: You must define your array in the `.data` segment and load its base address using the code skeleton below:

```
.data
A: .word 2, 13, 1, 4, 8, 20, 5, 6, 9, 14
K: .word 10

.text
.globl main
main:
    la $t1, A         # Load base address of A[0] into $t1
    lw $t2, K         # Load value of K into $t2

    # Your code goes here...

    li $v0, 10        # Exit simulator
    syscall
```

Testing Requirement: Test your code using **two different arrays** and at least **three different K values** for each array (minimum of 6 total test executions).

2) Function Calls and Stacks (50 Points)

Write a MIPS code for the following C code fragment.

Constraints:

- You are **NOT** allowed to use any multiplication instructions (`mult`, `multu`, or `mul`). Use addition or bit-shifts (`sll`) instead.
- You **MUST** use the `jal` instruction for function calls.
- You **MUST** properly manage register saving (including `$ra` and callee-saved registers) on the stack during nested function calls.

```c
int main() {
    int a;
    int b;
    int result = 0;

    if (a > b) {
        result = process_data(a, b);
    } else {
        result = (a + b) << 2; // Equivalent to 4 * (a + b)
    }
    return result;
}

int process_data(int a, int b) {
    if (a % 2 == 0) {
        return compute_alpha(a, b);
    } else {
        return compute_beta(a, b);
    }
}

int compute_alpha(int a, int b) {
    // Computes 4*a - b without using multiplication
    return ((a << 2) - b);
}

int compute_beta(int a, int b) {
    // Computes 2*a + 3*b without using multiplication
    return ((a << 1) + (b << 1) + b);
}
```

Testing Requirement: Test your program for all branches ($a > b$ with a even, $a > b$ with a odd, and $a \leq b$) using different integer values for a and b.

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What to Turn In

Please submit a **single PDF report** named `CMP634_yourname_lastname.pdf` via the Google Form submission link provided at the top of this document. If you resubmit before the deadline, only your latest submission will be evaluated.

Your PDF report must contain:

1. **Assembly Codes:** Paste your complete, commented MIPS `.asm` code directly into the document.
2. **Screenshots:** Provide clear screenshots of QtSpim/MARS showing register and memory states **before and after** each test run.

Grading Policy

- **Question 1 (Code & Tests):** 50 Points
- **Question 2 (Code & Tests):** 50 Points